

# Formal objects, morphisms, and glossary for the Type-1, $N = 3$ Landau–Zener reduction

shared geometry, selector data, and the continuation functor

This note formalizes the two diagrams in the strategy document. It does two things. First, it replaces each box by a named mathematical object. Second, it replaces each arrow by a named map, process, or partially defined morphism. The resulting diagram is meant to be inspected by a mathematician who has not seen the project before: the base ODE sits at the left, the shared spectral geometry in the middle, and the unresolved exact-WKB data are concentrated in a single broken morphism on the right.

The note builds on the augmented adiabatic-basis ODE, the quotient-ring reduction, the commuting triple  $(H_x, Y, R)$ , and the common-cover / selector framework developed in the summary and primer.

## 1 Foundational geometric objects inherited from the summary and primer

We fix a Type-1,  $N = 3$  commuting family, a member  $H$ , a preferred incoming channel  $x$ , and a Borel direction  $\theta$ . The family-invariant spectral data are the rational functions

$$q_u(\lambda) = u p(\lambda) - n(\lambda), \quad u(\lambda) = \frac{n(\lambda)}{p(\lambda)}, \quad E(\lambda) = \frac{m_H(\lambda)}{p(\lambda)}.$$

The gauged adiabatic amplitudes satisfy

$$f'(u) = [-S(u) + iD_H(u)]f(u),$$

where  $S$  is the transport connection of the moving adiabatic frame and  $D_H$  is diagonal in the adiabatic basis.

For  $N = 3$ , the squared pairwise gaps live on the cubic phase curve

$$\chi_Y(u, y) = y^3 - 2A(u)y^2 + A(u)^2y - D(u) = 0,$$

and the transport coordinate is

$$r(\lambda) = \frac{W_4(\lambda)}{L_H(\lambda)^4}.$$

The common phase–transport cover and its spinor lift are

$$\mathcal{C} = \Sigma_Y \times_{\mathbb{P}_u^1} \Sigma_R, \quad \hat{\mathcal{C}} = \{(u, y, r, \rho) \in \mathcal{C} \times \mathbb{C} : \rho^2 = \kappa_S/r\},$$

with global one-forms

$$\omega = \sqrt{y} du, \quad \eta = S du = \frac{\rho}{y} du.$$

The commuting triple is

$$(H_x, Y, R), \quad Y = M_{Q_4 L_H^2 / p^2}, \quad R = M_{W_4 / L_H^4}.$$

Away from the transport discriminant

$$\mathfrak{C}_R = \{u \in \mathbb{P}_u^1 : \text{Disc}_r \chi_R(u, r) = 0\},$$

the spectral projectors of  $R$  define the canonical splitting

$$\mathcal{E}_u = \mathcal{L}_1(u) \oplus \mathcal{L}_2(u) \oplus \mathcal{L}_3(u).$$

This is the precise sense in which gluing is already solved on the regular set.

## 2 Formal reduction objects

We now package the strategy into named objects.

**Definition 2.1** (Base ODE object). *The base ODE object is*

$$\mathbf{B}_{H,x} := (U, \mathcal{F}_{H,x}, \nabla_{H,x}^{\text{ad}}), \quad \mathcal{F}_{H,x} := U \times \mathbb{C}^3, \quad \nabla_{H,x}^{\text{ad}} := \partial_u + S(u) - \mathbf{i}D_H(u),$$

so that  $\nabla_{H,x}^{\text{ad}} f = 0$  is the gauged adiabatic-basis equation.

**Definition 2.2** (Reduced algebraic object). *The reduced algebraic object is*

$$\mathbf{Q}_{H,x} := (U, \mathcal{E}, \nabla_{H,x}^{\text{red}}; q_u, s_u), \quad \mathcal{E}_u := \mathbb{C}[\lambda]/(q_u(\lambda)), \quad \nabla_{H,x}^{\text{red}} := \partial_u - \mathbf{i}H_x,$$

where  $s_u$  is the inverse of  $p$  in  $\mathcal{E}_u$  and  $H_x = M_{h_x}$  with

$$h_x(\lambda; u) = \text{red}_{q_u}((m_H(\lambda) - E_x(u)p(\lambda))s_u(\lambda)).$$

**Definition 2.3** (Commuting algebra object). *The commuting algebra object is*

$$\mathbf{A}_H := (\mathcal{E}, H_x, \mathbf{Y}, \mathbf{R}),$$

where  $\mathbf{Y}$  and  $\mathbf{R}$  are the multiplication operators with symbols

$$\mathfrak{y}(\lambda) = \frac{Q_4(\lambda)L_H(\lambda)^2}{p(\lambda)^2}, \quad \mathfrak{r}(\lambda) = \frac{W_4(\lambda)}{L_H(\lambda)^4}.$$

**Definition 2.4** (Spectral geometry object). *The spectral geometry object is*

$$\mathbf{G}_H := (\Sigma_Y, \Sigma_R, \mathcal{C}, \widehat{\mathcal{C}}; \omega, \eta),$$

where

$$\Sigma_Y = \{(u, y) : \chi_Y(u, y) = 0\}, \quad \Sigma_R = \{(u, r) : \chi_R(u, r) = 0\}, \quad \mathcal{C} = \Sigma_Y \times_{\mathbb{P}_u^1} \Sigma_R, \quad \widehat{\mathcal{C}} = \{\rho^2 = \kappa_S/r\},$$

and  $\omega, \eta$  are the global phase and transport one-forms on  $\widehat{\mathcal{C}}$ .

**Definition 2.5** (Ramification object). *The ramification object is*

$$\mathbf{R}_H := (B_R, \mathfrak{C}_R, \pi_R),$$

where  $\pi_R : \Sigma_R \rightarrow \mathbb{P}_u^1$  is the projection,  $B_R := \text{Ram}(\pi_R)$  is its ramification divisor, and  $\mathfrak{C}_R := \pi_R(B_R)$  is the transport discriminant locus.

**Definition 2.6** (Candidate selector object). *The candidate selector object is*

$$\mathbf{S}_{H,\theta}^{\text{cand}} := (\text{Sel}_{\theta}^{\text{cand}}(H), \mathcal{S}_{+-}),$$

where

$$\text{Sel}_{\theta}^{\text{cand}}(H) := \{u_* \in \mathfrak{C}_R : (e^{-\mathbf{i}\theta} \mathcal{S}_{+-}(u_*; H)) = 0\}, \quad \mathcal{S}_{+-}(u) := \int_{\gamma_{+-}(u)} \omega.$$

**Definition 2.7** (Local activation–multiplier object). *The local activation–multiplier object is*

$$\mathbf{J}_{H,\theta} := (\text{Sel}_{\theta}^{\text{act}}(H), \{J_s(H)\}_{s \in \text{Sel}_{\theta}^{\text{act}}(H)}),$$

where  $\text{Sel}_{\theta}^{\text{act}}(H) \subseteq \text{Sel}_{\theta}^{\text{cand}}(H)$  is the actual selector subset and  $J_s(H) \in GL_3(\mathbb{C})$  is the local jump matrix attached to  $s$ .

**Remark 2.8.** *Object 2.7 is the only object in the chain that is not presently known by a closed intrinsic construction. Determining it is the content of the unresolved local exact-WKB problem.*

**Definition 2.9** (Decorated continuation graph). *The decorated continuation graph is*

$$\mathbf{G}_{H,\theta}^{\text{dec}} := (\mathfrak{G}_{H,\theta}, V, E, \ell),$$

where

$$V = \text{Ord}_\theta(H) \cup \text{Sel}_\theta^{\text{act}}(H) \cup V_\infty,$$

with  $\text{Ord}_\theta(H)$  the ordinary turning-point set and  $V_\infty$  the chosen asymptotic sector vertices. The edges are the Stokes and transport sectors between these vertices, and the label function  $\ell$  assigns propagation data to edges and local jump data to vertices.

**Definition 2.10** (Path groupoid and continuation functor). *The path groupoid object is*

$$\mathbf{P}_{H,\theta} := \Pi_1(\mathfrak{G}_{H,\theta}) = \Pi_1(\mathfrak{G}_{H,\theta}),$$

and the continuation functor object is

$$\mathbf{T}_{H,\theta} := (\Pi_1(\mathfrak{G}_{H,\theta}), \mathcal{T}_{H,\theta}), \quad \mathcal{T}_{H,\theta} : \Pi_1(\mathfrak{G}_{H,\theta}) \rightarrow GL_3(\mathbb{C}).$$

**Definition 2.11** (Output object). *The output object is*

$$\mathbf{O}_{H,\theta} := (f(+\infty), P), \quad P_{x \rightarrow j} = |f_j(+\infty)|^2.$$

### 3 Named morphisms and processes

The arrows in the category-style diagram are the following constructions.

**Definition 3.1** (Spectral reduction and interpolation). *The map*

$$\mathfrak{q}_{H,x} : \mathbf{B}_{H,x} \longrightarrow \mathbf{Q}_{H,x}$$

is defined by passing from the adiabatic coefficients  $f_i$  to the reduced sheet carriers  $R_i = f_i/\Gamma_i$ , and then to the unique interpolation class  $[R] \in \mathcal{E}_u$  with values  $R(\lambda_i(u); u) = R_i(u)$ .

**Definition 3.2** (Adjoining the commuting multipliers). *The map*

$$\mathfrak{a}_H : \mathbf{Q}_{H,x} \longrightarrow \mathbf{A}_H$$

keeps the same quotient-ring bundle and connection and adjoins the multiplication operators  $\mathbf{Y}$  and  $\mathbf{R}$ .

**Definition 3.3** (Spectral-curve construction). *The map*

$$\mathfrak{s}_H : \mathbf{A}_H \longrightarrow \mathbf{G}_H$$

assigns to the commuting algebra its spectral curves  $\Sigma_Y = \text{Spec}(\mathbf{Y})$  and  $\Sigma_R = \text{Spec}(\mathbf{R})$ , their fiber product  $\mathcal{C}$ , the spinor lift  $\widehat{\mathcal{C}}$ , and the one-forms  $\omega, \eta$ .

**Definition 3.4** (Ramification extraction). *The map*

$$\mathfrak{b}_H : \mathbf{G}_H \longrightarrow \mathbf{R}_H$$

extracts the transport ramification divisor of the projection  $\pi_R : \Sigma_R \rightarrow \mathbb{P}_u^1$  and its image in the base.

**Definition 3.5** (Phase-resonance sieve). *The map*

$$\mathfrak{c}_{H,\theta} : \mathbf{G}_H \longrightarrow \mathbf{S}_{H,\theta}^{\text{cand}}$$

is the function

$$\mathbf{G}_H \longmapsto \left( \left\{ u_* \in \mathfrak{C}_R : (e^{-i\theta} \mathcal{S}_{+-}(u_*, H)) = 0 \right\}, \mathcal{S}_{+-} \right).$$

It is the formal version of “transport ramification plus phase resonance.”

**Definition 3.6** (Local activation–multiplier morphism). *The partially defined morphism*

$$\mathfrak{j}_{H,\theta} : \mathbf{S}_{H,\theta}^{\text{cand}} \dashrightarrow \mathbf{J}_{H,\theta}$$

assigns to the candidate selector object the actual active subset and its local jump matrices.

**Remark 3.7.** *The broken arrow notation in the diagram is intentional:  $\mathfrak{j}_{H,\theta}$  is the unresolved local analytic input. Every other arrow in the reduction is already formalized.*

**Definition 3.8** (Graph decoration). *The map*

$$\mathfrak{d}_{H,\theta} : (\mathbf{G}_H, \mathbf{J}_{H,\theta}) \longrightarrow \mathbf{G}_{H,\theta}^{\text{dec}}$$

builds the decorated continuation graph by adjoining the ordinary turning set, the active selector set, and the corresponding edge/vertex labels.

**Definition 3.9** (Fundamental-groupoid construction). *The map*

$$\mathfrak{p}_{H,\theta} : \mathbf{G}_{H,\theta}^{\text{dec}} \longrightarrow \mathbf{P}_{H,\theta}$$

assigns to the decorated graph its path groupoid.

**Definition 3.10** (Ordered-product continuation functor). *The map*

$$\mathfrak{t}_{H,\theta} : \mathbf{P}_{H,\theta} \longrightarrow \mathbf{T}_{H,\theta}$$

assigns to each morphism class in  $\Pi_1(\mathfrak{G}_{H,\theta})$  the corresponding ordered product of edge propagators and local jump matrices. On generators,

$$\mathcal{T}_{H,\theta}(e) = D_e(H), \quad \mathcal{T}_{H,\theta}(s) = J_s(H),$$

and on composites it is multiplicative.

**Definition 3.11** (Evaluation). *For a chosen continuation word  $\gamma \in \Pi_1(\mathfrak{G}_{H,\theta})$ , the evaluation map*

$$\mathfrak{e}_{H,\theta,\gamma} : \mathbf{T}_{H,\theta} \longrightarrow \mathbf{O}_{H,\theta}$$

is given by

$$\mathfrak{e}_{H,\theta,\gamma}(\mathcal{T}_{H,\theta}) = (\mathcal{T}_{H,\theta}(\gamma)f(-\infty), |\mathcal{T}_{H,\theta}(\gamma)f(-\infty)|^2).$$

**Proposition 3.12** (Reduction up to the unresolved local step). *Once the local activation–multiplier morphism  $\mathfrak{j}_{H,\theta}$  is supplied, the long-time Landau–Zener amplitudes are determined functorially by the composition*

$$\mathbf{B}_{H,x} \xrightarrow{\mathfrak{q}_{H,x}} \mathbf{Q}_{H,x} \xrightarrow{\mathfrak{a}_H} \mathbf{A}_H \xrightarrow{\mathfrak{s}_H} \mathbf{G}_H \xrightarrow{\mathfrak{c}_{H,\theta}} \mathbf{S}_{H,\theta}^{\text{cand}} \xrightarrow{\mathfrak{j}_{H,\theta}} \mathbf{J}_{H,\theta} \xrightarrow{\mathfrak{d}_{H,\theta}} \mathbf{G}_{H,\theta}^{\text{dec}} \xrightarrow{\mathfrak{p}_{H,\theta}} \mathbf{P}_{H,\theta} \xrightarrow{\mathfrak{t}_{H,\theta}} \mathbf{T}_{H,\theta} \xrightarrow{\mathfrak{e}_{H,\theta,\gamma}} \mathbf{O}_{H,\theta}.$$

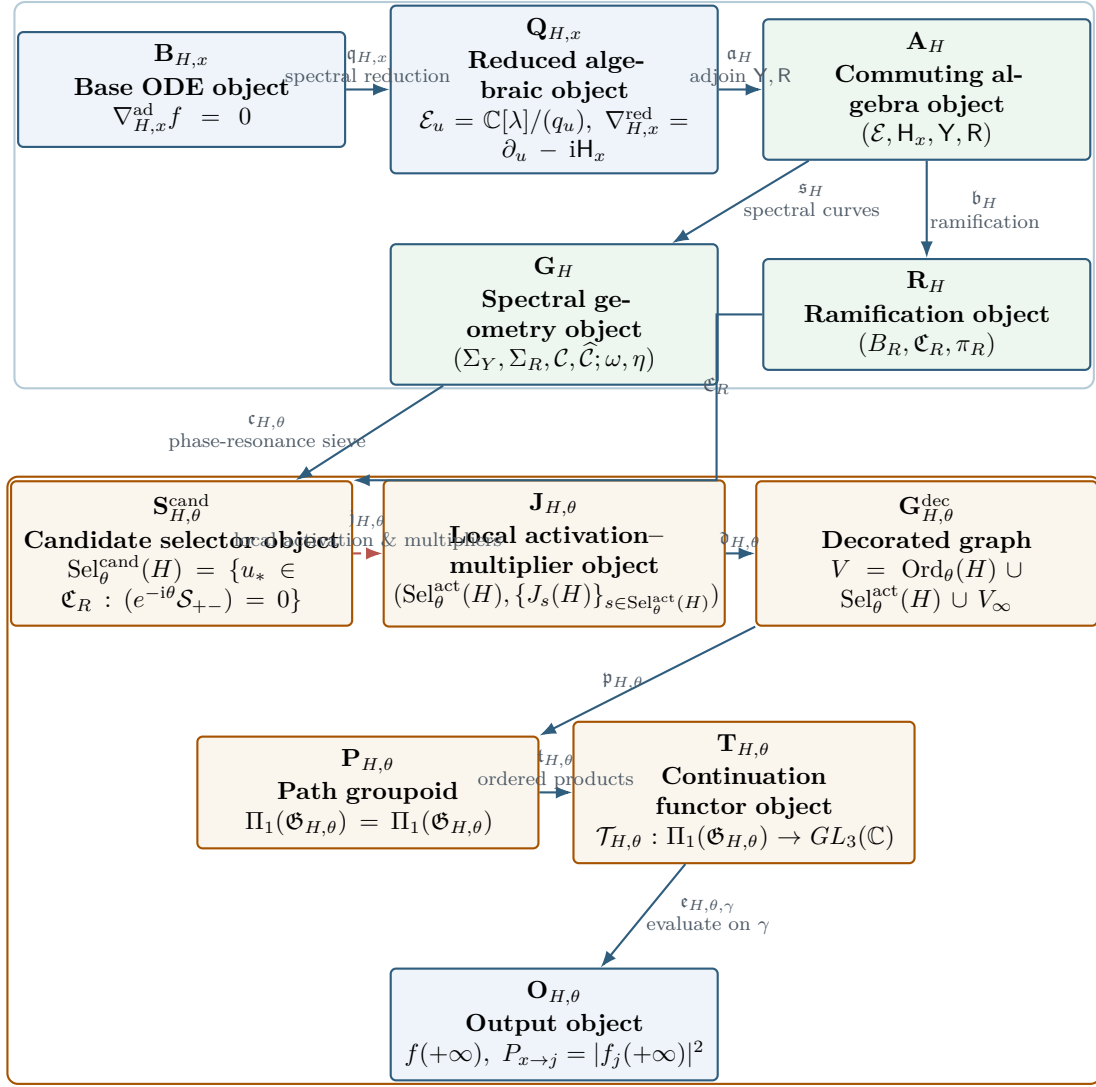


Figure 1: **Formal strategy diagram with named objects and arrows.** Every box is now a formally defined object. Every arrow is a named map or process from Section 3. The only broken arrow is  $j_{H,\theta}$ : the local activation–multiplier morphism that turns candidate selector points into active selector points equipped with jump matrices.

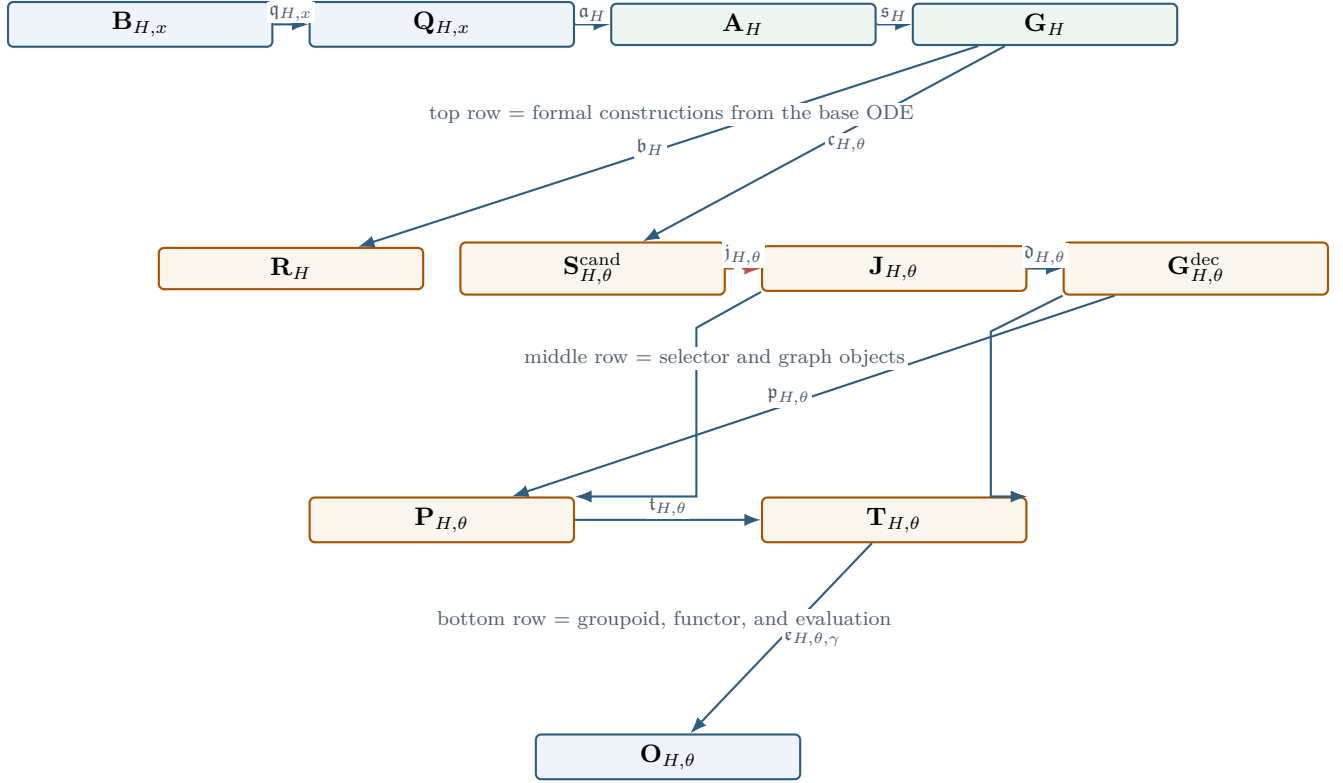


Figure 2: **Category-style reduction diagram.** The reduction is organized as a chain of named objects and named morphisms. The only non-formalized morphism is the dashed map  $j_{H,\theta}$ , which must decide which candidate selector points are genuinely active and what local jump matrices they carry.

## 4 Glossary of foundational objects (from the summary and primer, augmented)

| Symbol                  | Definition                                                                                    | Role in the reduction                                             |
|-------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| $q_u(\lambda)$          | $up(\lambda) - n(\lambda)$                                                                    | Spectral polynomial defining the quotient-ring fiber.             |
| $\mathcal{E}_u$         | $\mathbb{C}[\lambda]/(q_u)$                                                                   | Rank-three fiber algebra that packages the three spectral sheets. |
| $[R]$                   | Interpolation class in $\mathcal{E}_u$ with values $R(\lambda_i(u); u)$ at the roots of $q_u$ | Intrinsic reduced carrier.                                        |
| $H_x$                   | Multiplication by $h_x = \text{red}_{q_u}((m_H - E_x p)s_u)$                                  | Quotient-ring Hamiltonian for the reduced evolution.              |
| $Y$                     | Multiplication by $Q_4 L_H^2 / p^2$                                                           | Operator whose spectrum gives $y = \Delta^2$ .                    |
| $R$                     | Multiplication by $W_4 / L_H^4$                                                               | Operator whose spectrum gives the transport coordinate $r$ .      |
| $\Sigma_Y$              | $\{(u, y) : \chi_Y(u, y) = 0\}$                                                               | Global phase curve for squared gaps.                              |
| $\Sigma_R$              | $\{(u, r) : \chi_R(u, r) = 0\}$                                                               | Global transport curve.                                           |
| $\mathcal{C}$           | $\Sigma_Y \times_{\mathbb{P}_u^1} \Sigma_R$                                                   | Common phase–transport cover.                                     |
| $\widehat{\mathcal{C}}$ | $\{(u, y, r, \rho) : \rho^2 = \kappa_S / r\}$                                                 | Spinor lift resolving the transport square root.                  |

| Symbol                 | Definition                                                                      | Role in the reduction                                              |
|------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------|
| $\omega$               | $\sqrt{y} du$                                                                   | Phase differential used in action comparison.                      |
| $\eta$                 | $\rho du/y$                                                                     | Transport differential on the spinor lift.                         |
| $B_R$                  | $\text{Ram}(\pi_R)$ for $\pi_R : \Sigma_R \rightarrow \mathbb{P}_u^1$           | Transport-side branch locus upstairs.                              |
| $\mathfrak{C}_R$       | $\pi_R(B_R) = \{u : \text{Disc}_r \chi_R(u, r) = 0\}$                           | Transport discriminant locus downstairs.                           |
| $\mathcal{S}_{+-}$     | $\int_{\gamma_{+-}} \omega$                                                     | Phase mismatch between the two locally colliding transport sheets. |
| $\text{Ord}_\theta(H)$ | Ordinary turning-point set, equivalently the zero set of the phase discriminant | Ordinary Stokes vertices.                                          |

## 5 Glossary of strategy objects (new in this note)

| Object                                | Formal definition | Interpretation                                                                                                                |
|---------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------|
| $\mathbf{B}_{H,x}$                    | Definition 2.1    | The original gauged adiabatic Landau–Zener problem as a connection on a trivial rank-three bundle.                            |
| $\mathbf{Q}_{H,x}$                    | Definition 2.2    | The reduced quotient-ring formulation of the same dynamics.                                                                   |
| $\mathbf{A}_H$                        | Definition 2.3    | The commuting algebra $(\mathcal{E}, \mathbf{H}_x, \mathbf{Y}, \mathbf{R})$ that simultaneously controls phase and transport. |
| $\mathbf{G}_H$                        | Definition 2.4    | The spectral geometry extracted from the commuting algebra.                                                                   |
| $\mathbf{R}_H$                        | Definition 2.5    | The transport ramification data.                                                                                              |
| $\mathbf{S}_{H,\theta}^{\text{cand}}$ | Definition 2.6    | Ramification points filtered by phase resonance in Borel direction $\theta$ .                                                 |
| $\mathbf{J}_{H,\theta}$               | Definition 2.7    | The active selector subset together with its local jump matrices.                                                             |
| $\mathbf{G}_{H,\theta}^{\text{dec}}$  | Definition 2.9    | The continuation graph carrying ordinary and selector vertices and edge labels.                                               |
| $\mathbf{P}_{H,\theta}$               | Definition 2.10   | The path groupoid of the decorated continuation graph.                                                                        |
| $\mathbf{T}_{H,\theta}$               | Definition 2.10   | The matrix-valued continuation functor.                                                                                       |
| $\mathbf{O}_{H,\theta}$               | Definition 2.11   | The long-time amplitudes and probabilities.                                                                                   |

## 6 Glossary of named morphisms and their status

| Morph./process       | Domain             | Codomain           | Meaning / status                                                                     |
|----------------------|--------------------|--------------------|--------------------------------------------------------------------------------------|
| $\mathfrak{q}_{H,x}$ | $\mathbf{B}_{H,x}$ | $\mathbf{Q}_{H,x}$ | Spectral reduction and interpolation. Already formalized.                            |
| $\mathfrak{a}_H$     | $\mathbf{Q}_{H,x}$ | $\mathbf{A}_H$     | Adjoin the commuting multipliers $\mathbf{Y}$ and $\mathbf{R}$ . Already formalized. |
| $\mathfrak{s}_H$     | $\mathbf{A}_H$     | $\mathbf{G}_H$     | Spectral-curve / fiber-product construction. Already formalized.                     |

| Morph./process                   | Domain                                  | Codomain                              | Meaning / status                                                                                                   |
|----------------------------------|-----------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| $\mathfrak{b}_H$                 | $\mathbf{G}_H$                          | $\mathbf{R}_H$                        | Extract transport ramification data. Already formalized.                                                           |
| $\mathfrak{c}_{H,\theta}$        | $\mathbf{G}_H$                          | $\mathbf{S}_{H,\theta}^{\text{cand}}$ | Phase-resonance sieve: transport ramification plus Stokes-real phase mismatch. Already formalized as a rule.       |
| $\mathfrak{j}_{H,\theta}$        | $\mathbf{S}_{H,\theta}^{\text{cand}}$   | $\mathbf{J}_{H,\theta}$               | Local activation and multiplier assignment. <b>Open.</b>                                                           |
| $\mathfrak{d}_{H,\theta}$        | $(\mathbf{G}_H, \mathbf{J}_{H,\theta})$ | $\mathbf{G}_{H,\theta}^{\text{dec}}$  | Decorate the graph by ordinary-selector vertices and local labels. Formal once $\mathfrak{j}_{H,\theta}$ is known. |
| $\mathfrak{p}_{H,\theta}$        | $\mathbf{G}_{H,\theta}^{\text{dec}}$    | $\mathbf{P}_{H,\theta}$               | Fundamental-groupoid construction. Standard.                                                                       |
| $\mathfrak{t}_{H,\theta}$        | $\mathbf{P}_{H,\theta}$                 | $\mathbf{T}_{H,\theta}$               | Ordered-product continuation functor. Formal once local jumps are known.                                           |
| $\mathfrak{e}_{H,\theta,\gamma}$ | $\mathbf{T}_{H,\theta}$                 | $\mathbf{O}_{H,\theta}$               | Evaluate the continuation functor on a chosen continuation word and incoming state. Standard.                      |

## 7 A concise reduction principle

**Proposition 7.1** (What is solved, and what is left). *The formal reduction of the Type-1,  $N = 3$  Landau–Zener problem is complete up to the single broken morphism  $\mathfrak{j}_{H,\theta}$ . Equivalently:*

- *the passage from the base ODE to the commuting algebra is formalized;*
- *the passage from the commuting algebra to the common cover, spinor lift, and transport ramification locus is formalized;*
- *the candidate selector set is formalized by the phase-resonance sieve;*
- *the only genuinely missing analytic datum is the map that turns candidate selector points into active selector points carrying local jump matrices.*

*Once  $\mathfrak{j}_{H,\theta}$  is known, the continuation functor and therefore the long-time amplitudes are determined by functorial assembly.*

**Remark 7.2.** *This is the sharpest sense in which the problem is now “solvable up to step 5.” The geometry, the algebra, the ramification, and the candidate-selector sieve are already formalized. The unresolved content is concentrated in a single local exact-WKB assignment.*